On the Geometric Asymmetry of Functional Income Distribution: A Formal Note

0. Motivation and Construction

In the income approach to GDP, the capital and labor components at year i are C(i) and L(i) respectively. Define the auxiliary series $\phi(i) = C(i)/L(i)$ and its inverse $\lambda(i) = L(i)/C(i)$.

Let $\phi^* = \min(\phi(i))$ and $M = \max(\lambda(i)) = 1/\phi^*$ denote the historical extremes: M is the year of maximum relative advantage of Labor, ϕ^* the year of maximum relative advantage of Capital.

Construction of S1A

Fix year i with current capital share C(i). Had the distributional conditions of year M prevailed today, Labor would have received $M \cdot C(i)$ instead of its actual $L(i) = \lambda(i) \cdot C(i)$. The absolute gain of Capital — equivalently, the absolute loss of Labor relative to its best historical conditions — is therefore:

$\Delta K(i) = (M - \lambda(i)) \cdot C(i)$

To express this as a *relative* gain, we normalize not by today's economy C(i) + L(i), but by the counterfactual economy as it would have been today under the conditions most favorable to Capital (worst for Labor):

$GDP^* = C(i) + M \cdot C(i) = (M + 1) \cdot C(i)$

Dividing $\Delta K(i)$ by GDP^* and simplifying by C(i):

$S1A(i) = (M - \lambda(i)) / (M + 1)$

S1A is thus a counterfactual relative measure: the gain of Capital at year i, expressed as a fraction of the economy as it would have been under maximum Labor disadvantage. It is independent of the absolute scale of the economy, since C(i) cancels exactly.

Construction of S1B

The construction follows symmetrically, exchanging the roles of Capital and Labor: the reference year is now ϕ^* (maximum advantage of Capital, minimum of Labor), and the counterfactual denominator is the economy as it would have been under maximum Capital disadvantage. This yields:

*$S1B(i) = (\phi(i) - \phi) / (1 + \phi)$***

Setting $r = \phi(i)$, both series can be written compactly as:

$S1A = (r - \phi^*) / r(1 + \phi^*)$; $S1B = (r - \phi^*) / (1 + \phi^*)$; $S1A = (1/r) \cdot S1B$

The empirical anchor ϕ^* corresponds to historically meaningful years: USA 1970, UK, Italy and Spain 1976, France 1982 — the peak of the postwar labor-capital compromise in each country, preceding the neoliberal reorientation of the 1980s.

1. Asymptotic properties

1. $\lim(r \to 0)$ S1A = $-\infty$
2. $\lim(r \to \infty)$ S1A = $1/(1+\phi^*)$ ← finite upper bound
3. $\lim(r \to 0)$ S1B = $-\phi^*/(1+\phi^*)$
4. $\lim(r \to \infty)$ S1B = $+\infty$ ← unbounded

S1A is bounded above by construction; S1B is unbounded. The biological survival constraint is the only effective limit on S1B.

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2. The divergence (the wedge)

$|S1B - S1A| = |(r-1)|(r-\phi^*) / r(1+\phi^*)$

The sign depends exclusively on $(r-1)$:

- $r > 1$ (as in Italy) : $S1B - S1A > 0$ → Labor loses more than Capital gains
- $r = 1$: $S1B - S1A = 0$ → both series vanish simultaneously
- $r < 1$: $S1A - S1B > 0$ → Capital loses more than Labor gains

Convexity of the wedge:

$f(r) = (r-\phi^*)(r-1)/r = r - (1+\phi^*) + \phi^*/r$

$f''(r) = 2\phi^*/r^3 > 0$ for all $r > 0$, all $\phi^* > 0$

The accelerating divergence is a structural property of the functional form, independent of the historical parameter ϕ^*.

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3. First derivatives

$d(S1A)/dr = \phi^* / r^2(1+\phi^*)$ → decreasing in r, tends to 0

$d(S1B)/dr = 1/(1+\phi^*)$ → constant, independent of r

The ratio: $[d(S1A)/dr] / [d(S1B)/dr] = \phi^*/r^2 \to 0$ as $r \to \infty$

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4. Second derivatives

$\partial^2 S1A/\partial r^2 = -2\phi^* / r^3(1+\phi^*) < 0$ → strictly concave

$\partial^2 S1B/\partial r^2 = 0$ → linear

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5. Three structural correspondences with Marx

5.1 — The General Law of Capitalist Accumulation (polarization)

Marx (*Capital*, Vol. I, Ch. 25): *"Accumulation of wealth at one pole is, therefore, at the same time accumulation of misery, agony of toil, slavery, ignorance, brutality, mental degradation, at the opposite pole."*

The wedge $(r-1)(r-\phi^*)/r(1+\phi^*)$ is a theorem, not a metaphor. For $r > 1$, polarization is convex and accelerating in r, for any value of the historical parameter ϕ^*.

5.2 — The Tendency of the Rate of Profit to Fall (satiation of Capital)

Marx (*Capital*, Vol. III, Ch. 13): the progressive difficulty of extracting proportional surplus value as the system expands.

$\partial^2 S1A/\partial r^2 = -2\phi^*/r^3(1+\phi^*) < 0$: the marginal relative gain of Capital is strictly decreasing for all r and all ϕ^*. The deceleration is structural.

5.3 — The Biological Exhaustion of Labor Power (the "Vampire")

Marx (*Capital*, Vol. I, Ch. 10): Capital *"takes no account of the health and the length of life of the worker"* and acts as a vampire with no intrinsic limit, stopping only when the host organism is exhausted.

$d(S1B)/dr = 1/(1+\phi^*)$: the marginal extraction rate is constant — it does not decelerate with increasing r. The only effective brake is the biological constraint (demographic collapse).

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6. Summary

The geometry of S1A and S1B encodes a universal asymmetry: the party in distributional disadvantage faces potentially unlimited relative losses, while the party in advantage faces a structurally bounded relative gain. This wedge not only exists but accelerates convexly with distributional imbalance — for any ϕ^*, for any country. The three core tendencies identified by Marx in *Capital* emerge as analytical properties of normalized national accounts series.

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